

DB LAB 10 Submission

Name: Yash Raj

Roll Number: 24k-0737

Task01:

```
SET SERVEROUTPUT ON;
```

```
DECLARE
```

```
    base_price NUMBER := 250;
```

```
    tax NUMBER;
```

```
    total NUMBER;
```

```
BEGIN
```

```
    tax := base_price * 0.15;
```

```
    total := base_price + tax;
```

```
    DBMS_OUTPUT.PUT_LINE('Tax: ' || tax);
```

```
    DBMS_OUTPUT.PUT_LINE('Total Price: ' || total);
```

```
END;
```

```
/
```

Output:

```
PL/SQL procedure successfully completed.
```

```
Tax: 37.5
```

```
Total Price: 287.5
```

```
PL/SQL procedure successfully completed.
```

Task02:

```
SET SERVEROUTPUT ON;
```

```
DECLARE
```

```
  v_name VARCHAR2(50);
```

```
  v_salary NUMBER;
```

```
BEGIN
```

```
  SELECT first_name, salary
```

```
  INTO v_name, v_salary
```

```
  FROM employees
```

```
  WHERE employee_id = 105;
```

```
  DBMS_OUTPUT.PUT_LINE('Name: ' || v_name);
```

```
  DBMS_OUTPUT.PUT_LINE('Salary: ' || v_salary);
```

```
END;
```

```
/
```

Output:

```
Name: David
```

```
Salary: 4800
```

```
PL/SQL procedure successfully completed.
```

Task03:

```
SET SERVEROUTPUT ON;
```

```
BEGIN
```

```
  UPDATE employees
```

```
  SET salary =
```

```
    CASE
```

```
      WHEN salary < 8000 THEN salary + 1200
```

```
      ELSE salary + 400
```

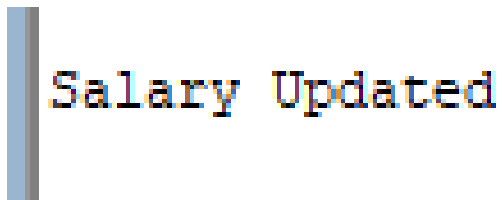
```
    END
```

```
  WHERE employee_id = 110;
```

```
  DBMS_OUTPUT.PUT_LINE('Salary Updated');
```

```
END;
```

```
/
```

Output:

```
Salary Updated
```

Task04:

```
SET SERVEROUTPUT ON;
```

```
DECLARE
```

```
  v_dept NUMBER;
```

```
  v_label VARCHAR2(50);
```

```
BEGIN
```

```
  SELECT department_id INTO v_dept
```

```
  FROM employees
```

```
  WHERE employee_id = 115;
```

```
  IF v_dept = 90 THEN
```

```
    v_label := 'Administration';
```

```
  ELSIF v_dept = 60 THEN
```

```
    v_label := 'Technical Services';
```

```
  ELSE
```

```
    v_label := 'General Staff';
```

```
  END IF;
```

```
  DBMS_OUTPUT.PUT_LINE('Department: ' || v_label);
```

```
END;
```

```
/
```

Output:

```
Department: General Staff
```

Task05:

```
DECLARE
v_salary NUMBER;
v_comm NUMBER;
v_bonus NUMBER;
BEGIN
SELECT salary, commission_pct
INTO v_salary, v_comm
FROM employees
WHERE department_id = 90
AND ROWNUM = 1; -- sample one employee

IF v_salary BETWEEN 15000 AND 20000 THEN
    v_bonus := v_salary * v_comm;
ELSIF v_salary > 20000 THEN
    v_bonus := (v_salary * v_comm) + 1000;
ELSE
    v_bonus := 0;
END IF;

DBMS_OUTPUT.PUT_LINE('Bonus: ' || v_bonus);
END;
/
```

Output:

```
Bonus:
```

```
PL/SQL procedure successfully completed.
```

Task06:

SET SERVEROUTPUT ON;

BEGIN

FOR emp IN (

SELECT first_name, salary

FROM employees

WHERE department_id = 50

)

LOOP

DBMS_OUTPUT.PUT_LINE(emp.first_name || ' - ' || emp.salary);

END LOOP;

END;

/

Output:

Matthew - 8000			
Adam - 8200	James - 2500	Trenna - 3500	Julia - 3400
Payam - 7900	TJ - 2100	Curtis - 3100	Anthony - 3000
Shanta - 6500	Jason - 3300	Randall - 2600	Kelly - 3800
Kevin - 5800	Michael - 2900	Peter - 2500	Jennifer - 3600
Julia - 3200	Ki - 2400	Winston - 3200	Timothy - 2900
Irene - 2700	Hazel - 2200	Jean - 3100	Randall - 2500
James - 2400	Renske - 3600	Martha - 2500	Sarah - 4000
Steven - 2200	Stephen - 3200	Girard - 2800	Britney - 3900
Laura - 3300	John - 2700	Nandita - 4200	Samuel - 3200
Mozhe - 2800	Joshua - 2500	Alexis - 4100	Vance - 2800
James - 2500	Trenna - 3500	Julia - 3400	Alana - 3100
Vance - 2800			
Alana - 3100			
Kevin - 3000			
Donald - 2600			
Douglas - 2600			

Task07:

```
SET SERVEROUTPUT ON;
```

```
CREATE OR REPLACE PROCEDURE get_salary(p_id NUMBER)
```

```
IS
```

```
  v_salary NUMBER;
```

```
BEGIN
```

```
  SELECT salary INTO v_salary
```

```
  FROM employees
```

```
  WHERE employee_id = p_id;
```

```
  DBMS_OUTPUT.PUT_LINE('Salary: ' || v_salary);
```

```
END;
```

```
/
```

```
SET SERVEROUTPUT ON;
```

```
BEGIN
```

```
  get_salary(101);
```

```
END;
```

```
/
```

Output:

```
Procedure GET_SALARY compiled
```

```
Salary: 17000
```

Task08:

```
SET SERVEROUTPUT ON;
```

```
CREATE OR REPLACE FUNCTION get_dept(p_id NUMBER)
```

```
RETURN NUMBER
```

```
IS
```

```
  v_dept NUMBER;
```

```
BEGIN
```

```
  SELECT department_id INTO v_dept
```

```
  FROM employees
```

```
  WHERE employee_id = p_id;
```

```
  RETURN v_dept;
```

```
END;
```

```
/
```

```
SET SERVEROUTPUT ON;
```

```
DECLARE
```

```
  v_dept NUMBER;
```

```
BEGIN
```

```
  v_dept := get_dept(101);
```

```
  DBMS_OUTPUT.PUT_LINE('Dept: ' || v_dept);
```

```
END;
```

```
/
```

Output:

```
Dept: 90
```


Task09:

```
SET SERVEROUTPUT ON;
```

```
CREATE OR REPLACE PROCEDURE increase_val(p_val IN OUT NUMBER)
```

```
IS
```

```
BEGIN
```

```
    p_val := p_val + (p_val * 0.25);
```

```
END;
```

```
/
```

```
SET SERVEROUTPUT ON;
```

```
DECLARE
```

```
    v_num NUMBER := 100;
```

```
BEGIN
```

```
    increase_val(v_num);
```

```
    DBMS_OUTPUT.PUT_LINE('Updated: ' || v_num);
```

```
END;
```

```
/
```

Output:

```
Updated: 125
```

Task10:

SET SERVEROUTPUT ON;

```
CREATE OR REPLACE PROCEDURE add_branch(  
    p_street VARCHAR2,  
    p_city VARCHAR2  
)
```

IS

```
    v_id NUMBER;
```

BEGIN

```
    SELECT NVL(MAX(branch_id), 0) + 1 INTO v_id  
    FROM branches;
```

```
    INSERT INTO branches(branch_id, street, city)  
    VALUES (v_id, p_street, p_city);
```

```
    DBMS_OUTPUT.PUT_LINE('Branch Added with ID: ' || v_id);
```

END;

/